

Orbit Decay AI Engine

Research & Implementation Methods

Project Watchdog

Research Methods

To accurately predict orbital decay, we analyzed:

- **Historical Aggregation:** Mapped UWE-4 TLE data against CelesTrak SW-All.csv solar weather data.
- **Physics Modeling:** Integrated NRLMSISE-00 for atmospheric density calculation.
- **Feature Engineering:**

- 7-Day & 30-Day F10.7 Solar Flux rolling averages.
- Kp Max geomagnetic activity indices.
- Ballistic coefficient (B^*) extraction.
- **Machine Learning:** Trained Random Forest regressors for precise drag predictions.

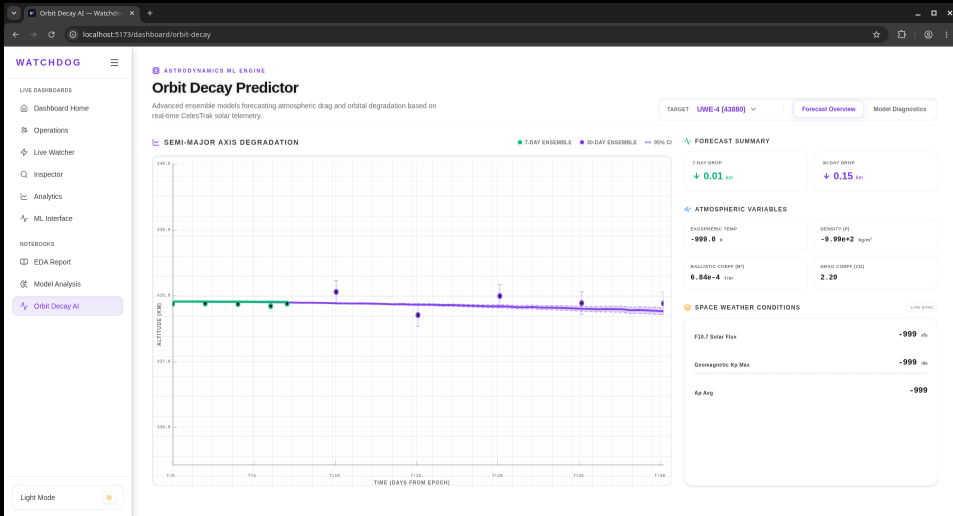
Deployment & Resiliency

The AI models have been integrated into our production stack:

- **FastAPI Endpoint:** Provides on-demand 7D and 30D forecast inferences.
- **Memory Caching:** All CelesTrak outgoing API calls are wrapped in an aggressive 3600s @ttl_cache to prevent rate-limiting.

- **Environment Locking:** Strict adherence to `scikit-learn==1.6.1` across Pixi and Docker to prevent deserialization faults.
- **Svelte UI:** A seamless, reactive dashboard utilizing WebSockets and interactive charts.

Results: Forecast Overview



Results: Model Diagnostics

